

Measuring the Association Between Short-Term Rental Supply and Long-Term Rental Prices in Calgary

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This research paper was produced as part of the City of Calgary–Urban Alliance Agreement made effective on 12 January 2023 (Research Services File Number 1060024). The City of Calgary is collaborating with researchers at the University of Calgary (UCalgary), under the Urban Alliance partnership, on a multi-year study of Calgary's short-term rental (STR) market. The goals of the study are to build a comprehensive evidence base on Calgary's STR market (including impacts, challenges, and opportunities), and to use this information to recommend a tailor-made STR policy framework for Calgary that can be adapted as market conditions change. The authors thank the City of Calgary for funding this work.

Executive Summary

Public belief and economic theory suggest that as the supply of short-term rental (STRs) increases, the prices (e.g., rent) of long-term rentals (LTRs) increases. This has been a difficult effect to conclusively empirically measure as omitted variable problems plague causal models and the effect may be empirically different dependent on the specific geography and unique STR market being looked at. As more cities, including Calgary, continue to grapple with whether and how STRs should be regulated, understanding the STR market's effect on the housing market for permanent residents is essential to create appropriate, evidence-informed regulations. In this paper, we measure the association between the increase in the STR supply and LTR prices in the city of Calgary, Alberta over the time period 2015 to 2022.

To measure the association between STR supply and LTR rents, we use Airbnb and VRBO data scrapped by AirDNA and LTR rent data collected by Canada Housing and Mortgage Corporation (CMHC) and a regression analysis at the community-year level. It should be noted that these are not causal results, e.g., we cannot conclude whether an increase in STRs *cause* an increase in LTR rents in Calgary. Our result takes-aways are:



In the City of Calgary, changes in STR listings are not statistically significantly associated with a change in average LTR rents.



In the City of Calgary, the magnitude of the effect of STRs on LTR rents is small. An increase of 100 STRs is associated with an increase of average LTR rents by 1.6 to 6.6 percent.



There is heterogeneity by community but even in the communities with the most STR activity, the magnitude of the association between STR and LTR rents is small. The increase of STRs in the Beltline community between 2017 and 2022—the community with the most STRs in Calgary—is associated with an increase of LTR rents by \$34/month, explaining 15 percent of the increase in LTR rents in the Beltline (with the remaining 85 percent explained by factors other than STRs).

Table of Contents

Authors of this report.....	1
Executive Summary	3
1. Introduction	5
2. Data and Descriptive Statistics	6
2.1 STR Data.....	6
2.2 LTR Rent Data	7
2.3 Creating the Final Data set	8
2.4 Descriptive Statistics.....	8
3. Empirical Strategy	11
4. Results	12
Conclusion	18
References.....	19

1. Introduction

Public belief and economic theory suggest that as the supply of short-term rental (STRs) increases, the prices (e.g., rent) of long-term rentals (LTRs) increases. This has been a difficult effect to conclusively empirically measure as omitted variable problems plague causal models and the effect may be empirically different dependent on the specific geography and unique STR market being looked at. As more cities, including Calgary, continue to grapple with whether and how STRs should be regulated, understanding the STR market's effect on the housing market for permanent residents is essential to create appropriate, evidence-informed regulations. In this paper, we measure the association between the increase in the STR supply and LTR prices in the city of Calgary, Alberta over the time period 2015 to 2022.

To measure the association between STR supply and LTR prices we use Airbnb and VRBO data scrapped by AirDNA and LTR rent data collected by Canada Housing and Mortgage Corporation (CMHC). Using a regression analysis at the community-year level, we find that a change in STR listings is not statistically significantly associated with a change in LTR rent and that the magnitude of the association is small. That is, our results suggest that an increase of 100 STRs in Calgary is associated with an increase of average LTR rents by between 1.6 and 6.6 percent but this is not statistically significant across multiple regression specifications. Our results also suggest that there is heterogeneity across communities. The increase of STRs in the Beltline community between 2017 and 2022—the community with the most STRs in Calgary—is associated with an increase of LTR rents by \$34/month, explaining 15 percent of the increase in LTR rents in the Beltline (with the remaining 85 percent explained by factors other than STRs). It should be noted that these are not causal results, e.g., we cannot conclude whether an increase in STRs *cause* an increase in LTR rents in Calgary.

Economic theory suggests that the effect of an increase in STR supply has an ambiguous effect on LTR rents. Consider a standard downward sloping demand curve and an upward sloping supply curve for the LTR market. On the one hand, an increase in STR supply decreases LTR supply, shifting the supply curve for LTRs to the left, increasing the LTR price—a substitution effect. On the other hand, it is also possible that an increase in the STR supply in a neighborhood may decrease the demand curve for LTRs: less people will demand an LTR in a neighborhood with a lot of noisy, nuisance STRs—a negative externality. This shifts the demand curve for LTRs to the left, potentially decreasing LTR rents. The final effect on LTR rents is dependent on the slope of the LTR demand and supply curve as well as the magnitude of the shifts. For example, if both demand and supply are sloping (e.g., are not horizontal/vertical), and the substitution effect is larger than the negative externality, LTR prices will increase. However, if the negative externality is larger than the substitution effect, it is possible that LTR rents decrease. The empirical literature attempts to measure these shifts.

The Canadian empirical literature on the association between STR supply and LTR prices is sparse. Using a mixed effects regression model and scraped Airbnb data, Wachsmuth (2023) measures the association between the share of frequently-rented entire home/apartment (FREH) STRs and the average LTR rent in a neighborhood in BC from 2016 to 2021. He finds that an increase of one FREH per 100 households is associated with an increase in the average LTR rent by \$0.49 (and this increase is statistically significant).¹ Likewise, using data directly from Airbnb, The Conference Board of Canada (2023) uses a fixed-effects model to measure the association between the share of high-use STRs on the log of average LTR rent. They find that a one percentage point increase in the share of high-use STRs increases average LTR rents by 0.2 percent, but this result is not statistically significant. It is

¹ Note that the Wachsmuth (2023) paper incorrectly reports the magnitude of the association between STR supply and LTR prices by a factor of 100. We have corrected it here.

important to note that these two papers use different measures of STRs (“frequently rented entire apartment/houses” per 100 households in Wachsmuth (2023) and “dedicated” STRs divided by number of households in the CBC report) and both likely suffer from omitted variable bias: changes in neighborhoods over time affect both the supply of STRs and LTR prices but is not captured in the models. Regardless, overall, the Canadian literature suggests that an increase in STRs that are more commercialized (e.g., frequently listed on the STR platforms and booked) is associated with either a small, statistically significant increase in LTR prices, or a small, not statistically significant increase in LTR prices.

Outside of Canada, a number of papers in the US and Europe examine the effect of Airbnb on the housing market prices. Barron, Kung, and Proserpio (2021) examine the effect of Airbnb’s on LTR rents and house prices for all of the US. Using an instrumental variables approach where the instrument is Google Trends Airbnb searches (time variation) combined with a neighborhood’s tourist index based on location of restaurants (cross-sectional variation), they find a one percent increase in Airbnb listings increases LTR rents by 1.8 percent—about one-fifth of actual rent growth. Koster, van Ommeren, and Volkhausen (2021) use a differences-in-differences design in Los Angeles to examine the effect of home sharing ordinances that severely restricted Airbnb’s in some cities and not others. They find that counties that restricted Airbnb’s saw their LTR rents drop by 2.3 percent and that this effect increased the closer one got to Hollywood (e.g., more touristy areas). Finally, examining Barcelona Spain, García López et al. (2020) use a number of econometric methods including a method similar to Barron, Kung, and Proserpio (2021) and find that an increase in 100 Airbnb’s in a neighborhood increased LTR rents by 1.9 percent, and, in high Airbnb areas, this effect increased to 17 percent. They argue that while Airbnb has taken off since 2013, it cannot explain the bulk of the high aggregate in rents the city has experienced. Overall, the international literature suggests that there is a positive causal effect of STRs on LTR rents. This effect is largest in areas that have higher concentrations of Airbnb’s and that are more “touristy”. Our results for Calgary are in line with this literature.

2. Data and Descriptive Statistics

2.1 STR Data

To measure STR activity, we use proprietary data on Airbnb and VRBO listings purchased from AirDNA for the city of Calgary. AirDNA scrapes data on STRs listed on Airbnb and VRBO/HomeAway. This scraped data includes two linkable data sets: the “property” data set that contains information on the characteristics of the listing itself (as reported by the host) such as the property type (e.g., apartment), listing type (e.g., entire apartment or single room), the number of bedrooms, and the published nightly fee. The second data set is the “monthly” data set. It contains aggregate monthly transaction data for each listing that includes the number of bookings per month, the number of days available in a month, the occupancy rate, and the monthly revenue. Unlike the property data, the monthly transaction data has had to be estimated by AirDNA from 2015 onwards as Airbnb and VRBO/HomeAway does not provide information on bookings. AirDNA uses machine learning algorithms based on an enormous historical dataset to perform the estimations. This introduces uncertainty; however, Calgary’s STR market is somewhat large so the number of observations in the historical dataset reduces the uncertainty. Outside of data supplied directly from STR platforms, we believe that this is the best available data set on the STR market.²

Before using this data, we cleaned it. Particularly, we removed listings posted by traditional tourist accommodations such as hotels, motels, lodges, inns, and hostels. While these may be listed on STR sites, they are

² This data is also widely used by other authors looking at STRs in Canada and/or the effects of STR regulation. See for examples: Combs, Kerrigan, and Wachsmuth (2020); DiNatale, Lewis, and Parker (2018).

not what we consider as STRs, which are accommodations that are in or on a property that could otherwise be used for residential purposes. Thus, we manually checked all hosts and listings in our data to ensure that traditional tourist accommodations were excluded from the analysis.

There are several other notes to be made with this data. First, if an Airbnb and VRBO/HomeAway listing share a calendar and are fundamentally the same listing, they are combined into a single STR listing by AirDNA. This avoids double-counting of identical STR listings. Second, listings are more proliferate than then number of private dwellings/properties used as an STR. Some hosts have multiple listings for the same private dwelling. For example, a host may have one listing for an entire single detached house, plus separate listings for separate rooms in the same house, plus separate listings for the upstairs and downstairs. Another common example is one host separately listing various rooms in one private dwelling. In both examples, there is only one private dwelling. This issue cannot be resolved with precision and must be borne in mind in the interpretation of results. Finally, some hosts use Airbnb not to list STR's but to list dwellings that they intend to rent for more than 28 days (i.e., a medium- or long-term rental). While we have removed those listings that, on our manual check indicated they were long-term, it is unclear the proportion of listings that are used in this manner.

The AirDNA data is spans January 2015 to September 2023. To match the LTR data which is only collected for the month of October in any given year (discussed below), we use only the monthly data for the month of October from 2015 to 2022.

2.2 LTR Rent Data

For data on LTR rents and LTR vacancies, we use data from the annual rental market survey conducted and collected by the CMHC.³ The CMHC conducts the rental market survey once per year in the month of October. It covers row houses and apartments that are part of the primary rental market—privately initiated structures with at least three rental units that have been on the market for at least three months. Social and affordable housing is not included. Rent is the amount actually paid by the tenant, or, in vacant units, it is the amount being asked for by the owner. No adjustments are made for the inclusion or exclusion of amenities and services such as heat, hydro, parking or hot water.

Specifically, we use data from the CMHC's annual rental market survey at the community- year level. The data spans the years 2010 to 2023 (inclusive; for the month of October only). To get the data at the community-year level, we contacted CMHC who created for us a customized data set.⁴

Because of the strict definition for “rentals” used by CMHC and because many communities do not have any or a sufficient number of “rentals” as defined and surveyed by the CMHC, many community-years are missing LTR rent data. In Calgary, there are 219 residential communities, excluding industrial communities, parks, and residual communities. Table 1 shows the number of residential communities per year for which CMHC has (non-missing) LTR rent data that make up our final data set (discussed below). Overall, there are 72 unique communities in the final data set, 42 (58 percent) of which have data available for all eight years from 2015 - 2022. Four communities only have data available for one year: Seton, Livingston, Red Carpet, and Panorama Hills.

³ At the time of this writing, the best available data on LTR rents in Calgary (and Canada broadly) comes from the CMHC rental market survey. Other similar research has used this CMHC data (e.g., Conference Board of Canada 2023; Wachsmuth 2023).

⁴ The CMHC rental market survey data is publicly available at the “neighborhood”-year level. A neighborhood spans multiple communities and does not line up with community boundaries—it is based on census tracts.

Table 1: Number of Communities with Non-Missing LTR Rent Observations by Year

Year	Number of Communities	As a Percent of Residential Communities
2015	61	27.9
2016	55	25.1
2017	52	23.7
2018	64	29.2
2019	57	26
2020	59	26.9
2021	63	28.8
2022	64	29.2

2.3 Creating the Final Data set

The final data set is at the community-year level. In creating the final data set we only use residential communities that have non-missing LTR rent data (note: there is no missing data for STRs). We supplement the LTR and STR data with single residential assessment data (available for all residential communities), school enrollment data (available for all communities where one or more schools is located), and geographical data where we measure the distance from the centre of the community to City Centre.⁵ Overall, for the months of October from 2015 to 2022 inclusive we have 475 community-year observations.

2.4 Descriptive Statistics

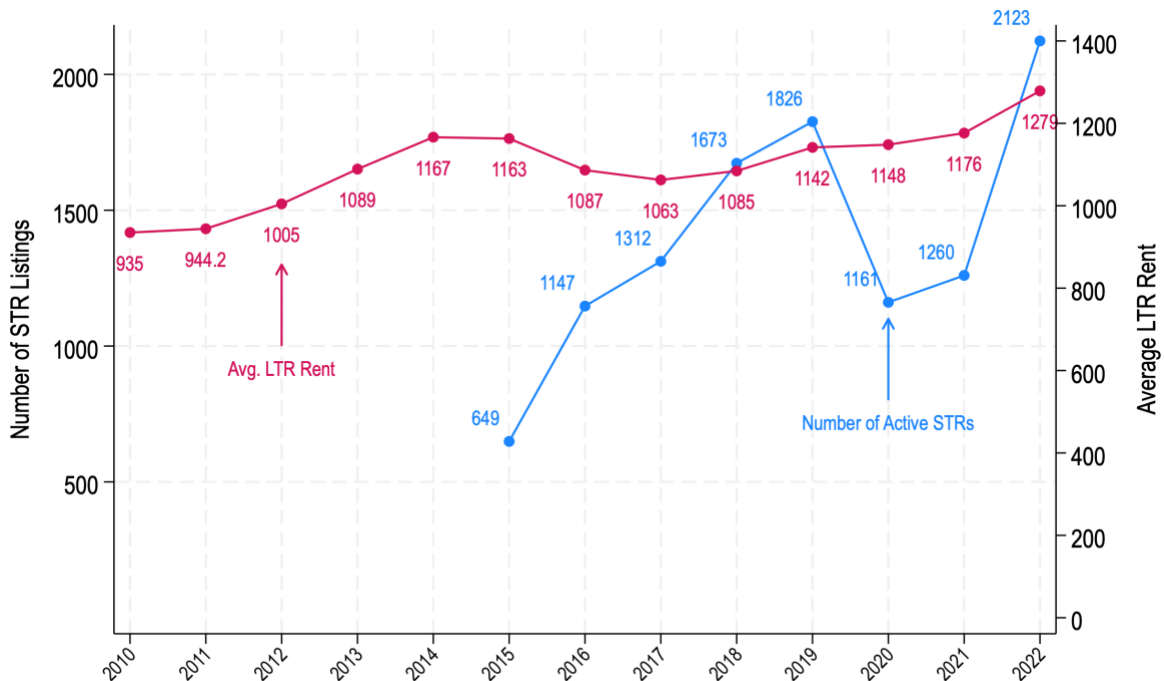
In Figure 1, we plot the number of active Airbnb and VRBO's (collectively "STR") listings in the month of October for the years 2010 to 2022 together with that of the average LTR rent for Calgary in October for 2015 to 2022 (for those community-years with non-missing data). The left-hand y-axis is for the number of STR listings, and the right-hand side y-axis is for LTR rent (in dollars). Active STR listings grew strongly pre-COVID-19 pandemic, increasing from 649 listings in October 2015 to 1,826 listings in October 2019. There was large growth in active STR listings between October 2015 and October 2016. This is potentially due to declining oil and gas prices between mid 2014 and early 2016 (Stocker, John, and Vorisek 2018) that resulted in mass layoffs (Geddes 2015) and gutted Calgary's downtown core (Bloomberg News 2015). Condo prices also fell during this time (McMahon 2015), making converting LTRs to STRs appealing to avoid large capital losses and empty LTRs. STR listings continued to increase until the COVID-19 pandemic. During the COVID-19 pandemic, active STR listings decreased from 1,826 listings in October 2019 to 1,161 and 1,260 listings in October 2020 and October 2021, respectively, before re-surging again in October 2022 up to 2,123 listings.

Over this same time, the average LTR rent did not follow a similar pattern. In October 2015, the average LTR rent was \$1,163/month. Rents declined in 2016 and 2017 as the effect of declining oil prices set in: mass layoffs in the oil and gas sector likely decreased demand for LTRs. After 2017, average LTR rents began increasing again from \$1,085/month in October 2018 to \$1,279/month in October 2022. There was no COVID-related decline in LTR rents. It should also be noted that while currently the housing crisis and affordable housing are top of mind for

⁵ The community boundaries, assessment data and school enrollment data come from the City of Calgary's Open Data website: <https://data.calgary.ca/>

Calgarians in 2024, the housing crisis began in Calgary in late 2022/2023, following on the heels of historically high inflation, a sharp increase in the bank rate, and a surge of immigration (Osuji and Scruggs 2023).

Figure 1: Number of STRs vs. LTR Rents, 2010 - 2022, Calgary



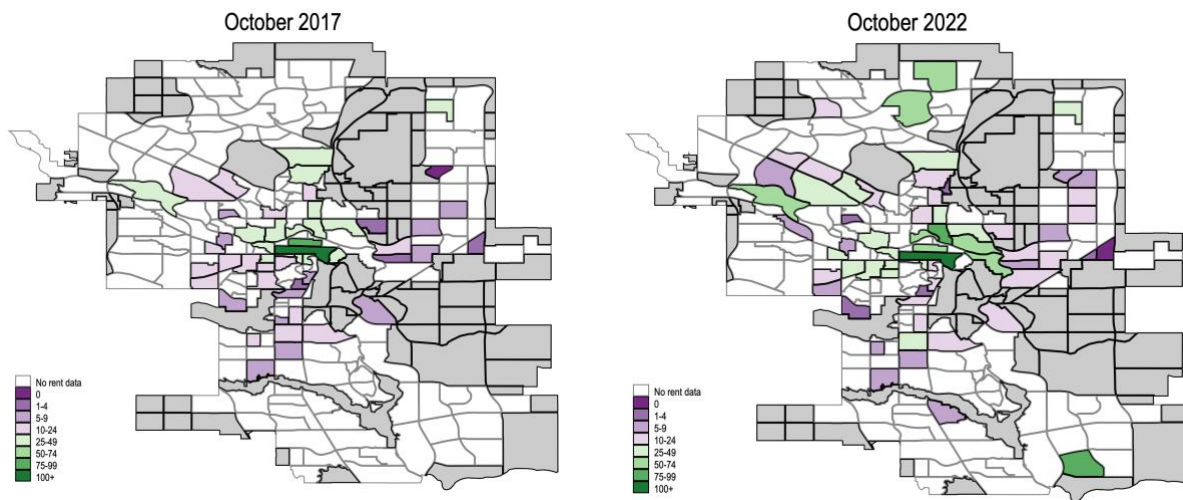
Data sources: CMHC Monthly Rental Survey for Calgary (average LTR rent); AirDNA (active STR listings).

There is spatial variation in STR activity in Calgary. Figure 2 shows the distribution of STRs by neighborhood in October 2017 and in October 2022.⁶ The gray areas are parks, industrial areas, residual sub-areas, and the airport: they are non-residential and thus do not have LTRs nor STRs. The communities that are color-coded white in Figure 2 are communities that do not have LTR rent data either because they do not have rentals covered by the CMHC survey or because they do not have sufficient number of rentals so the rental data is suppressed.

Figure 2 shows that of the communities with LTR rent data, those with the most STRs are in the downtown core and developing areas. The Beltline had the most STRs (483) in October 2022, followed by Crescent Heights (81) and Sunnyside with the fourth most (720—all in the Centre sector. Seton and Livingston—both developing communities in 2022—had 79 and 70 STRs respectively. As discussed in Petit and Tedds (2023), these areas have distinct STR types. The downtown core STRs tend to be entire apartments whereas STRs in the developing areas are more likely to be entire suites (where the permanent resident/homeowner/renter is more likely to also occupy the same property as their personal residence compared to entire apartments).

⁶ In Figure 2, we have chosen to compare 2017 and 2022 (instead of 2015 and 2022) because of what was occurring in the local economy. 2015 and 2016 were unusual years for the LTR and STR market as oil prices declined. We do not have STR data prior to 2015. The next “normal” year appears to be 2017. Further, 2022 is post-pandemic (although it is the start of the housing crisis).

FIGURE 2: NUMBER OF STRs BY CMHC NEIGHBORHOOD, CALGARY, OCTOBER 2017 AND OCTOBER 2022



In Table 2 we report descriptive statistics. The data is presented for two points in time: October 2017 (during the rise of the STR market in Calgary) and October 2022 (the current height of the STR market in Calgary). The data is also presented for two different sample: all neighborhoods and the top decile of neighborhoods. The top decile are neighborhoods in the top 10 percent of neighborhoods by STR active listings in October 2022.

TABLE 2: DESCRIPTIVE STATISTICS

	October 2017		October 2022	
	All Communities	Top STR Decile (By # of STRs)	All Communities	Top STR Decile (By # of STRs)
Avg. Active STR count per community	25	80	33	131
Avg. LTRs in the CMHC Survey per community	656	1,761	651	2,235
Avg. Number of STRs as a Percent of Residential Assessments per community	0.95%	1.45%	1.09%	2.40%
Avg. LTR Rent per community	\$1,063	\$1,093	\$1,279	\$1,339

Note: The STR top decile is determined by the number of STRs per community (given they have rent data) within each year.

Table 2 shows that while the average number of active STRs by neighborhood has grown over time, the growth has been particularly high in the top decile of communities. In the top decile of communities, the average number of active STRs grew by 64 percent (from an average of 80 listings to 131 listings) from October 2017 to October 2022 while the overall average growth was only 32 percent (from an average of 25 listings to 33 listings). The average of STR's as a percent of residential assessments has also increased. Across all communities with non-missing data, STRs increased from 0.95 percent of residential assessments to 1.09 percent of residential assessments—an increase of 15 percent. In the top decile of neighborhoods, STRs as a percent of residential assessments grew from 1.45 percent to 2.40 percent — a 66 percent increase.

Table 2 also shows that between October 2017 and October 2022, the average LTR rent increased. by \$216/month (20 percent) across all communities. In communities that are in the top decile by STR, average LTR rent increased

by \$246/month (22.5 percent): the increase in the average LTR rent was slightly larger in top decile neighborhoods. It is unclear if, and how much, of this increase is associated with STR activity, and is what this paper addresses.

3. Empirical Strategy

Before setting out the empirical specification, it is important to note that what we are trying to estimate is shifts in the demand and supply curves, e.g., how a change in STR supply shifts LTR supply and demand. It is very difficult to estimate a causal effect in this type of scenario. The main problem is that while we hypothesize an increase in the supply of STR's increases LTR rents, there are a multitude of other changes happening in the LTR market that we may be picking up: it is difficult to estimate the causal effect resulting from one specific change without better data and a clever empirical design.⁷ Associations are more straightforward to estimate but are not causal and thus cannot say with certainty whether STRs *cause* a change in the LTR market.

Regardless, following García López et al. (2020), the main analysis consists of estimating the following fixed-effects regression:

$$\log(LTR\ Rent)_{jt} = \beta(NumberOfSTRs)_{jt} + \phi X_{jt} + \tau_t + \mu_j + \varepsilon_{jt} \quad (1)$$

where *LTR rent* is the average LTR rent measured in October of year *t* for community *j*. *Number of STRs* is the number of active Airbnbs and VRBOs in in October of year *t* for community *j*. τ_t is year fixed effects. μ_j are time-invariant community fixed effects. The main concern with identification is that communities with the most STR activity might be concurrently experiencing sociodemographic changes or revitalization that has an impact on both the number of STRs and LTR rents. One way we control for this is to introduce time-varying community controls, X_{jt} . Demographic data at the community-year level is not available.⁸ Thus, to control for revitalization and/or gentrification, we use annual secondary school enrollments, the value of residential property assessments, and the number of residential property assessments.⁹ Controlling for the number of assessments and the number of secondary school enrollments controls proxies for population growth and growth of younger families. Controlling for the assessment value controls for changes in housing values over time in a community. Housing values may change due to market forces but also due to revitalization as older houses are replaced with newer houses.

We also run models that allow for different communities to have different time trends in the X_{jt} variables. To do this, we add as additional regressors $X_{jt}t$ in the linear case, and $X_{jt=2015}t$ and $X_{jt=2015}t^2$ in the quadratic case, where X_{jt} is as measured in 2015. In these models, we also include an interaction term between time trend and distance to city centre, allowing for neighborhoods closer to the centre to have steeper time trends.

Finally, we run models that allow each neighborhood to have different time trends in LTR rents. In these models, we add $\sum_j \rho_j t$ in the linear case, and $\sum_j \rho_j t + \sum_j \psi_j t^2$ in the quadratic case.

⁷ In Canada, we only (currently) have rental data for one month of the year (October) and aggregated up to a geographical area (e.g., community). Causal effects can be measured in other countries because there are data sets of individual LTR listings and their rental price (e.g., Idealista in Spain).

⁸ The best available data is Census data. While demographics as reported on the Census are available at the community level, the Census only occurs twice in our data sample: once in 2016 and again in 2021. This would limit the sample to only two years, thus it was not used.

⁹ All of this data comes from the Open Data Calgary portal and is publicly available.

In the case that STRs affect not only trends but levels of LTR rents, the following specification is not appropriate – it captures both the effects of STR and neighborhood specific time trends (Wolfers 2006). Garcia-Lopez et al. (2020) suggest a detrending method which we also use here. In particular, we run two regressions:

$$\log (Rent)_{jt} = \tau_t + \mu_j + \sum_j \rho_j t + \varepsilon_{jt} ; \text{ for } 2010 < t < 2015 \quad (2)$$

$$\log (\widetilde{Rent})_{jt} = \beta (Number\ of\ STRs)_{jt} + \varphi X_{jt} + \tau_t + \varepsilon_{jt} ; \text{ for } t > 2015 \quad (3)$$

In the first stage (equation 2), the log of rent is regressed on a time fixed effect, a neighborhood fixed effect, and a time trend that is allowed to vary by neighborhood only for the years pre-STR growth in Calgary (i.e., 2010 to 2015). We then use the estimated coefficients from that regression to predict the log of rents for all years (2010-2022) and the residuals. The residuals (that have been de-trended) are then regressed in the second regression (equation 3) using the baseline specification (equation 1).

4. Results

Table 3 reports the regression results. From Table 3, we see that regardless of the model used, the number of STRs in a neighborhood does not have a large effect on the average long-term rent in a neighborhood. In column 1—the model with only neighborhood and time fixed effects—the regression suggests that if active STR listings in a neighborhood increase by 100, LTR rents will increase by 4.3 percent (statistically significant at 90 percent). When neighborhood time-varying controls are added into the model in column 2, this declines to 1.8 percent and the result is no longer statistically significant.

In column 3 and 4 we include linear and quadratic control-specific time trends based on the level of controls in 2015, respectively. The magnitude of the estimated coefficient on the number of STR's in columns 3 and 4 is between the estimated coefficients in columns 1 and 2, suggesting consistent results. However, in both columns 3 and 4, the estimated coefficient on the number of STRs is not statistically significant. Columns 5 and 6 include a linear and quadratic time trend by neighborhood to allow for both observable and unobservable time trends to impact the average LTR rent. Again, the magnitude of the estimated coefficient on the number of STR's in column 5 is between the estimated coefficients in columns 1 and 2, suggesting consistent results while the magnitude of the coefficient is very small in column 6, suggesting that as STRs increase by 100, LTR rents increase by less than one percent. In both columns 5 and 6 the estimated coefficient on the number of STRs are not statistically significant.

Finally, columns 7 and 8 report the de-trending results using the method discussed above. That is, if the number of STRs affect not just levels by trends in LTR rents, by including the community fixed effect, this effect could be absorbing this effect. The magnitude of the coefficient in column 7 suggests that as STRs increase by 100, LTR rents increase by 1.6 percent whereas the coefficient in column 8 is the largest of all estimated coefficients and suggests that an increase of 100 STR listings increases the average LTR rent by 6.6 percent. The estimated coefficients in columns 7 and 8 are not statistically significant. Overall, our regression results suggest that an increase in the supply of STRs in Calgary has neither a statistically significant nor large (in magnitude) association with STR rents.

TABLE 3: REGRESSION RESULTS. INDEPENDENT VARIABLE = $\ln(\text{LTR RENTS})$

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Baseline	Baseline, Controls	Linear TT Controls	Quad TT Controls	Linear TT Controls & By Community	Quad TT Controls and by Community	De-Trended Baseline	De-Trended Baseline with Controls
Number of STRs (x 100)	0.0431** (0.0150)	0.0184 (0.0234)	0.0212 (0.0213)	0.0260 (0.0227)	0.0193 (0.0168)	0.000779 (0.0216)	0.0164 (0.0190)	0.0660 (0.0363)
Ln Assess. Value		0.214* (0.0955)	0.218* (0.0953)	0.224* (0.104)	0.0989 (0.126)	0.101 (0.100)		0.262 (0.147)
# Assessments		0.0000315 (0.0000326)	0.0000731* (0.0000347)	0.0000896* (0.0000436)	0.0000104 (0.0000335)	0.000131 (0.000120)		-0.0000877 (0.0000483)
# School Enroll		0.00000209 (0.00000614)	0.00000209 (0.00000645)	0.000000645 (0.00000526)	-0.00000556 (0.00000426)	-0.00000693* (0.00000316)		-0.00000939 (0.00000908)
Ln Assess. Value Linear TT			0.0108 (0.00809)	0.00386 (0.0208)	-0.00522* (0.00228)	-0.00377 (0.00924)		
# Assessments Linear TT			-0.00000353* (0.00000150)	-0.00000466 (0.00000498)	0.0000152*** (0.00000142)	0.000152*** (0.00000464)		
# School Enroll Linear TT			0.00000338 (0.00000274)	-0.00000327 (0.00000921)	-0.0000421*** (0.00000406)	-0.000307*** (0.0000164)		
Distance from Centre Linear TT			-0.115* (0.0500)	0.0440 (0.212)	-0.0587*** (0.00831)	-2.962*** (0.0847)		
Ln Assess. Value Quad TT				0.000815 (0.00243)		-0.000175 (0.00121)		
# Assessments Quad TT				7.55e-08 (0.000000507)		-0.0000157*** (0.000000548)		
# School Enroll Quad TT				0.000000760 (0.00000124)		0.0000305*** (0.00000206)		
Distance from Centre Quad TT				-0.0170 (0.0243)		0.322*** (0.0100)		
Community FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	475	475	440	440	440	440	475	475
R2	0.362	0.377	0.420	0.424	0.636	0.772	0.839	0.843
R2 Within	0.362	0.377	0.420	0.424	0.636	0.772	0.839	0.843
R2 Overall	0.157	0.118	0.0928	0.0962	0.202	0.157	0.280	0.193
R2 Between	0.301	0.131	0.0742	0.0813	0.123	0.138	0.0370	0.00865
# of Clusters	72	72	61	61	61	61	72	72
F	18.86	13.78	14.10	13.26	.	.	83.18	87.43

Note: Independent var: log of LTR rents. Run using xtreg with fe option and se's clustered on community in Stata 18. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

To understand the economic magnitude of these estimates, we focus on the results in column 2 (e.g., the baseline model with controls). Column 2 suggested that an increase of 100 STRs is associated with an increase in average LTR rents by 1.8 percent. The average increase in the number of STR listings in Calgary between 2017 and 2022 was 8 STR listings (see Table 2). Thus, these estimates imply an average increase of LTR rents by 0.14 percent. In the top decile neighborhoods, the average increase in active STR listings was 51 between 2017 and 2022. Thus, our estimates imply that in these neighborhoods, an average increase of LTR rents by 0.92 percent is associated with STR activity.

There is heterogeneity across communities. Taking the Beltline community for example (e.g., the neighborhood with the most STR activity), the average LTR rent in October 2017 was \$1,270/month and it increased by \$231 to \$1,501/month in October 2022. Over this same time, active STR listings in the Beltline increased by 147. Our estimates suggest that STR activity is associated with a 2.7 percent increase in average LTR rents in the Beltline or \$34/month, leaving the remaining \$197 of the increase to be explained by other factors. This can be compared to a community like Bowness where between 2017 and 2022, the average LTR rent increased by \$160 (from \$1,005 to \$1,165) and STR listings increased by 22 (from 30 to 52). Our model estimates that this increase in STR's is associated with an increase of the average LTR rent in Bowness by 0.396 percent or \$0.63/month, leaving the remaining \$159.37/month increase to be explained by other factors.

We show this heterogeneity in Figure 3. In panel A, we plot the percentage increase in LTR rents that our model suggests is associated with the increase in the number of active STR listings in a neighborhood between 2017 and 2022 (using column 2 estimates). In panel B, we translate this increase into dollar values. Finally, in Panel C, we plot the percentage of the actual LTR rental increase that can be explained by STRs. Note that these estimates can only be done for communities where we have non-missing LTR rents in both October 2017 and October 2022.

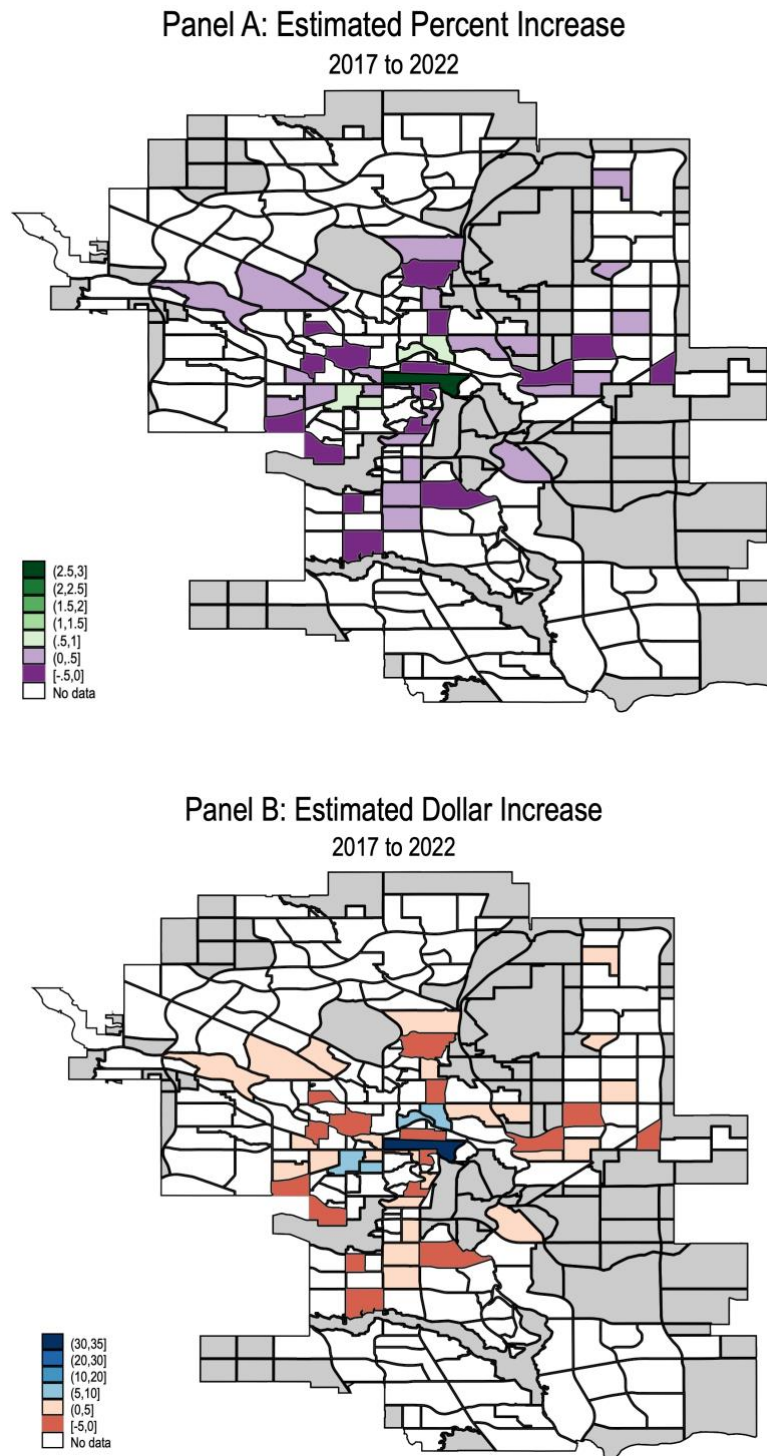
In Panel A, we see that our model estimates an association of between -0.5 percent and 3 percent. As discussed, our estimates suggest the community where the association between STR and LTR rents is the largest is the Beltline: an increase of STRs in the Beltline increased LTR rents by 2.7 percent. The communities of Sunnyside, Richmond, Crescent Heights, and South Calgary all also saw between a 0.5 percent and one percent increase in LTR rents associated with their increase in STRs. Panel A also shows that some communities saw a decrease in the LTR rents. This is because in some of these communities there was a decline in STR listings between 2017 and 2022. In particular, the Downtown Commercial Core saw a decline in their STR by 19 listings. This was associated with a decline in the LTR rents in the Downtown Commercial Core by 0.3 percent.

From Panel B, these changes translate to between a \$5/month decline and \$35/month increase in LTR rents. Our estimates suggest that STR activity was associated with the largest dollar increase in average LTR rents in the Beltline as already discussed: an increase in STRs is associated with an increase in LTR rents of \$34/month between 2017 and 2022. The communities of Sunnyside, Richmond, Crescent Heights, and South Calgary all also saw between a \$5/month and \$8/month increase in LTR rents associated with their increases in STR listings. Panel B also shows that some communities saw a decrease in the LTR rents associated with declines in STR listings. The Downtown Commercial Core saw a decline in their LTR rents by about \$4/month associated with their decline in STR listings.

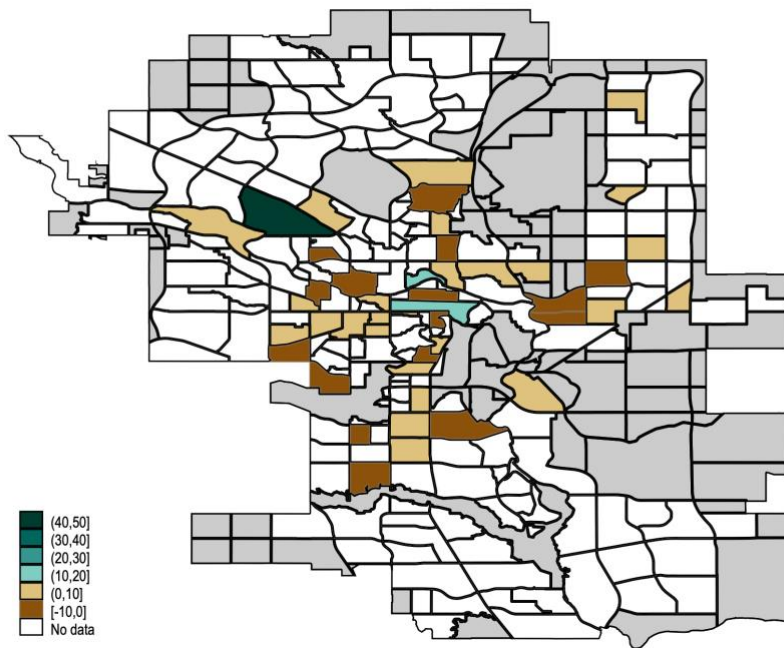
Finally, Panel C shows a somewhat different pattern. In the community of Varsity, our results suggest that an increase in STRs explains 47 percent of the actual increase in LTR rents in that community. However, in Varsity between 2017 and 2022, STR listings increased by 4 listings and LTR rents increased by \$7/month. Because of these small changes, our results suggest that STRs are associated with a \$1.9/month increase in LTR rents. In sum, while the percentage effect is large, the absolute magnitude is small.

Outside of Varsity, Panel C shows that the increase in STR listings in the Beltline can explain 15 percent of the increase in LTR rents in the Beltline. Likewise, in the community of Sunnyside, the increase in STR listings in the Beltline can explain 13 percent of the increase in LTR rents. At the other end of the spectrum, a decline in STR listings in the Downtown Commercial Core explains 2.5 percent of the decline in LTR rent in that community.

FIGURE 3: ESTIMATED ASSOCIATION BETWEEN STR LISTINGS AND AVG. LTR RENT BY NEIGHBORHOOD



Panel C: % Change in LTR Rent Explained by STR's
2017 to 2022



Overall, while our results suggest there is heterogeneity in the effect of STR listings on LTR rents by community, the effects are small in magnitude. Coupling this with our regression results, these effects are also not statistically significant.

Conclusion

In this paper we examined the association between STR listings and LTR rents in the City of Calgary. This is a difficult relationship to measure precisely, and it is even more difficult to determine causation (which we do not do here). However, our results do suggest that there is no significant association between STR listings and LTR rents in the City of Calgary. As the number of STR listings increase, there is no statistically significant nor economically significant correlated increase in LTR prices.

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